

**Title: Intelligent Resume Evaluation and Feedback System for Upskilling Using
AI**

A Synopsis Report of Major Project

**BACHELOR OF TECHNOLOGY
in
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Abstract

The proposed **Intelligent Resume Evaluation and Feedback System for Upskilling Using AI** is an AI-driven platform designed to automate resume evaluation while providing personalized feedback and career development recommendations to job seekers. Traditional Applicant Tracking Systems (ATS) primarily focus on filtering and ranking candidates based on predefined criteria, but they provide limited guidance regarding skill gaps and methods for improving employability.

The proposed system combines **Natural Language Processing (NLP)**, **Named Entity Recognition (NER)**, **semantic similarity analysis**, **Machine Learning**, and **Graph Neural Networks (GNNs)** to analyze resumes against job descriptions. The system extracts important information such as education, work experience, skills, certifications, and languages from resumes in formats such as PDF and DOCX. Semantic embeddings are then generated to enable context-aware comparison between candidate profiles and job requirements.

A graph-based matching mechanism using **Graph Convolutional Networks (GCN)** and **Relational Graph Convolutional Networks (RGCN)** is used to improve candidate-job matching and scoring. The system further performs skill-gap analysis and generates personalized recommendations for relevant courses, certifications, resume improvements, and career pathways.

The proposed system aims to transform resume screening from a passive filtering process into an active, feedback-driven learning process. The research evaluation reported an F1-score of **0.81** for the proposed GNN model, compared with 0.74 for BERT and 0.68 for TF-IDF, along with a **92% skill-gap detection accuracy** and an average user satisfaction rating of **4.6/5**.

Introduction

The modern recruitment process involves evaluating a large number of resumes against specific job requirements. Traditional resume screening is often performed manually or through Applicant Tracking Systems (ATS). Manual screening is time-consuming and may be influenced by subjective decisions, while conventional ATS solutions mainly depend on predefined keywords and criteria. This can result in qualified candidates being overlooked because their resumes do not contain exact keywords or appropriate formatting.

Artificial Intelligence, particularly **Natural Language Processing and Machine Learning**, provides an opportunity to automate and improve the resume evaluation process. AI-based systems can extract meaningful information from unstructured resumes, compare candidate profiles with job descriptions, and rank candidates according to their relevance.

The proposed system goes beyond conventional resume screening by providing candidates with actionable feedback. It identifies missing or underrepresented skills and recommends suitable learning resources, certifications, and career pathways. The system therefore provides benefits to both recruiters and candidates by improving candidate-job matching while helping candidates understand how they can improve their employability.

The major objectives of the proposed system are:

- To develop an automated **NLP-based resume parser** capable of extracting structured information from resumes.
- To compare resumes with job descriptions using **semantic similarity and ATS scoring**.
- To identify **skill gaps** between candidates and specific job requirements.
- To generate personalized recommendations for **courses, certifications, and learning paths**.
- To provide feedback related to **resume formatting, keyword optimization, and career development**.
- To improve candidate-job matching using **Graph Neural Networks**.
- To provide a user-oriented system that supports continuous resume improvement and career development.

Motivation

The recruitment process has become increasingly competitive, making it important for candidates to understand how well their skills and qualifications match the requirements of available jobs. Existing recruitment systems generally concentrate on helping recruiters filter and rank candidates, while providing limited assistance to candidates who want to understand why their profile may not match a particular job.

One of the major limitations of keyword-based resume screening is that it may fail to understand the context in which skills and qualifications are mentioned. It may also fail to recognize semantically similar terms and soft skills. In addition, most existing systems do not provide personalized guidance regarding which skills a candidate should develop.

The motivation behind this project is therefore to develop a system that not only evaluates a resume but also converts the evaluation into an **actionable upskilling plan**.

By combining NLP, semantic embeddings, and graph-based learning, the proposed system can provide more meaningful candidate-job comparisons.

The system is motivated by the following requirements:

- **Automated resume evaluation** to reduce manual effort.
- **Context-aware matching** instead of simple keyword matching.
- **Identification of skill deficiencies** according to job requirements.
- **Personalized feedback** for improving resumes.
- **Learning and certification recommendations** for identified skill gaps.
- **Transparent and growth-oriented recruitment support** for candidates.

Background

Artificial Intelligence has increasingly been incorporated into recruitment systems, particularly in the areas of resume parsing, candidate classification, and job matching. Natural Language Processing enables unstructured resume documents to be transformed into structured information such as education, experience, skills, and certifications.

Previous research has explored different approaches to resume analysis. NLP-based systems have been developed for extracting entities and comparing resumes with job descriptions. Machine Learning techniques such as **KNN, SVC, XGBoost, LSTM, and ensemble models** have also been used for resume classification and candidate ranking.

More advanced approaches use semantic embeddings and graph-based models. Graph Neural Networks can represent relationships between candidates, jobs, skills, education, and other attributes, allowing more complex relationships to be considered during candidate evaluation. The proposed system uses **GCN and RGCN** to model these relationships and improve resume-job matching.

The proposed system follows a multi-stage AI pipeline. Resumes and job descriptions are collected and pre-processed using normalization, tokenization, stop-word removal, and lemmatization. NER models such as **BERT, spaCy, and a fine-tuned RoBERTa-base model** are used for entity extraction. Domain-specific skills can additionally be identified using rule-based approaches such as FlashText.

The extracted entities are converted into semantic embeddings using models such as **Sentence-BERT** or **OpenAI's text-embedding-3-large**. These embeddings are compared with job requirements using cosine similarity. A graph is then constructed

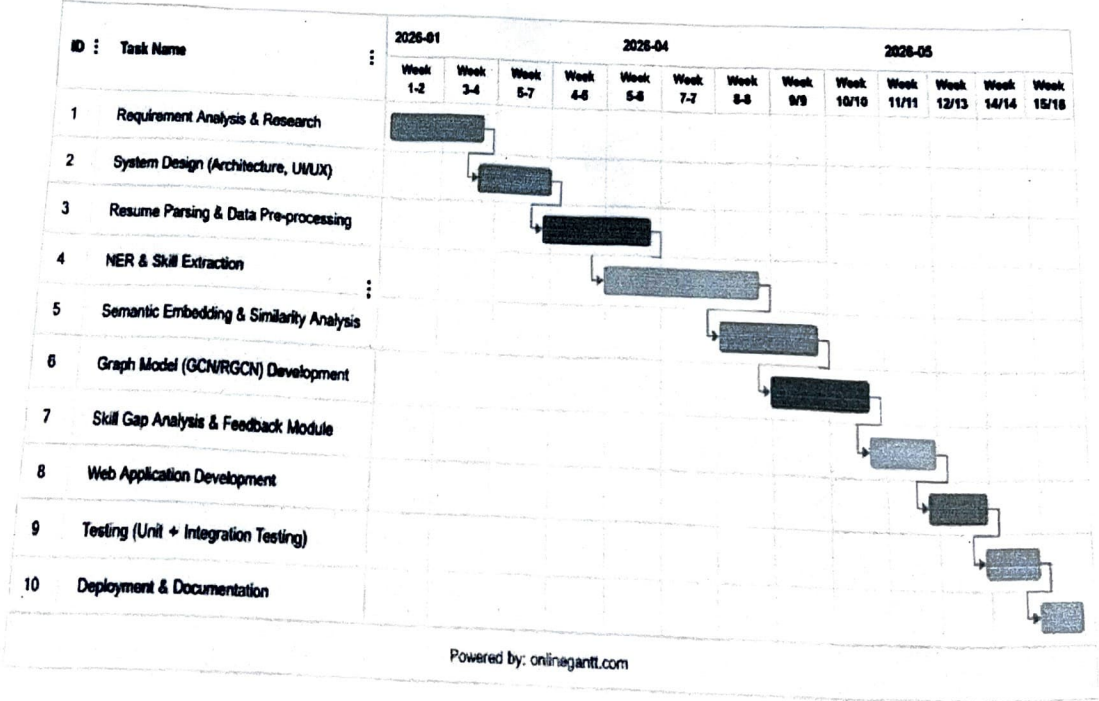
where candidates and their relationships are represented as nodes and edges, respectively. The proposed RGCN architecture uses three convolutional layers to capture multi-hop relationships.

Following resume-job matching, the system performs **skill-gap analysis** by identifying skills that are missing or underrepresented in the candidate's profile. Personalized feedback is then generated, including recommendations for online courses, certifications, resume improvements, and career development.

The major research gap identified is that existing systems largely focus on filtering and ranking candidates, while relatively few systems combine **resume parsing, semantic matching, skill-gap identification, personalized feedback, and upskilling recommendations** within a single framework. The proposed project aims to address this gap.

Planning of Work (Gantt Chart)

To ensure systematic progress, the following timeline is proposed for the project:



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